

1. Identification

Material Name : 2-Hydroxyethyl 2-methylacrylate, 2-Hydroxyethyl methacrylate

Synonyms : HOMA, HEMA

CAS Number : 868-77-9

Product Use : Raw material for coating resin, Paper coating resin, Adhesive, Resin modifier etc.

Manufacturer/Supplier : **Mitsubishi Chemical Corporation.**

Address : 1-1, Marunouchi 1-chome, Chiyoda-ku, Tokyo, Japan

Section : MMA domain (Chemicals)

Telephone number : +81-3-6748-7540

Fax number : +81-3-6685-0785

Emergency Telephone number : +81-3-6748-7540
(0900 – 1700 Japan office hours only).

2. Hazards Identification

GHS Classification:

Health	Environmental	Physical
Acute Toxicity - Not Classified (oral)	Hazardous to the aquatic environment (Acute)	Flammable Liquid
Acute Toxicity - Not Classified (dermal)	- Not Classified	- Not Classified
Eye Corrosion/Irritation - Category 2A	- Not Classified	
Skin Corrosion/Irritation - Not Classified	Hazardous to the aquatic environment (Long-term)	
Skin Sensitization - Category 1	- Not Classified	
Mutagenicity - Not Classified		
Reproductive/Developmental - Not Classified		

GHS Label:

Symbols:



Hazard Statements

WARNING

Causes severe eye irritant.

May cause allergic skin reaction.

Precautionary Statements

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood.

Wear protective gloves/protective clothing/eye protection/face protection/respiratory protection as required.

Keep away from heat/sparks/open flames/hot surfaces. - No smoking.

Keep container tightly closed.

Ground/bond container and receiving equipment.

Take precautionary measures against static discharge.

Use explosion-proof electrical/ventilating/lighting equipment.

Use only non-sparking tools.

Do not breathe mist/vapours.

Use only outdoors or in a well-ventilated area.

Avoid release to the environment.

Do not eat, drink or smoke when using this product.

Contaminated work clothing should not be allowed out of the workplace.

Wash hands thoroughly after handling.

Counter measures

IF INHALED: If you feel unwell or breathing is difficult, remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician.

IF ON SKIN: Remove/Take off immediately all contaminated clothing. Wash with plenty of soap and water. If you feel unwell or skin irritation or rash occurs, get medical advice/attention.

IF IN EYES: Rinse cautiously with water for over 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention.

IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician. Do NOT induce vomiting.

IF feeling unwell or experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician.

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IF exposed or concerned: Get medical advice/attention.

Collect spillage.

In case of fire, use dry chemical, carbon dioxide, foam, dry sand for extinction.

Storage

Storage in tightly closed container at cool and well-ventilated area.

Store locked up.

Dispose

Contents/container are burned in a designated incinerator or are entrusted to qualified waste trader.

Another Hazards out of GHS classification

This material is flammable at high temperature (flash point is 109°C). High temperature or direct rays may cause polymerization. Polymerization followed exothermic heat and may cause explosive reaction.

3. Composition / Information on Ingredients

Component	CAS Number	Weight %
2-Hydroxyethyl 2-methylacrylate	868-77-9	≥97.0%
EINECS Number	TSCA	
212-782-2	Registration	

4. First Aid Measures

Eye: Eye irritation. Flush immediately with large amounts of water for at least 15 minutes.

Eyelids should be held away from the eyeball to ensure thorough rinsing.

If possible, remove contact lenses before flushing the eyes.

Get immediate medical attention.

Skin: Immediately remove contaminated clothing and shoes. Flush the skin with plenty of running water at normal or lukewarm temperatures.

Get immediate medical attention if the appearance of the skin changes or if irritation persists.

Inhalation: Remove exposed person from source of exposure to fresh air. Wrap in a

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blanket or similar material to keep warm. Keep quiet. If breathing is weak or stops, loosen clothing, secure an open respiratory airway and start cardiopulmonary resuscitation (CPR).

Get immediate medical attention.

Ingestion: Get immediate medical attention. Do not induce vomiting unless directed by medical personnel. Doing so will increase the potential for harm. Should be allowed to wash the mouths with water, if the person himself wishes. Do not place anything in the mouth if the person is unconscious.

Most important Sign and Symptoms: In case inhalation, may cause dizziness or choking and a nose or the throat irritation. Contact to eye or skin may cause irritation and a burning by chemicals. If swallowed may cause feeling nausea, vomiting or griping.

Protection of rescuer: In case rescue the victims, wear suitable protective equipments.

5. Fire Fighting Measures

Suitable Extinguishing Media: Use dry chemical, foam, carbon dioxide or dry sand to extinguish fire.

Unsuitable Extinguishing Media: Do not use water in a jet. Doing so may be in danger the spread fire.

Special Hazards: Nothing special item.

Fire-fighting procedures: Keep persons not directly involved in fire-fighting out of the area of the fire. Use dry chemicals, a CO₂ extinguisher, dry sand etc. to extinguish a fire in its early stages. In case of large fires, deprive the fire of a source of oxygen using a foam extinguisher or by other means. Wear protective equipment during fire fighting.

6: Accidental Release Measures

Person protection measures (Protective equipments): Wear protective equipment and operate from upwind during clean up works. Isolate hazard area by rope and deny entry to unnecessary of unprotected personnel.

Environment protection measures: Do not discharge the spill into sewer pipes. If the spill involves small amounts, soak up the material with sawdust, waste cloth or sand, then collect the soaked materials into a container. For extensive spills, block the flow of liquid with earth or sand, cover the liquid surface with foam, then collect as much of the material into an empty container as possible. We recommend using spark proof tools.

Protects from secondary accidents: Promptly remove any flammable materials from

the surrounding area. Prepare extinguishing media providing ignition.

7. Handling and Storage

Handling:

- Do not use compressed air to charge, bleed or handle the product.
- If the product is to be handled indoors, provide good ventilation using a local exhaust system or other measures. (In case evaporating at 20 °C atmosphere may be contaminated and the concentration may arrive at poisonous level quickly.)
- Keep the product away from flames, sparks, strong oxidizers and high temperature substances.
- Handle the container gently; do not overturn, drop, strike or drag.
- Wear adequate protective equipment to prevent absorption through the skin, contact with the skin or eyes or inhalation of vapors. -- See Protective equipment.

Storage:

Keep container tightly closed. Store in a well-ventilated place. Keep cool. Store locked up. Keep away from heat, sparks, open flame, hot surfaces - No smoking. Protect from sunlight. Do not store in the same area with organic peroxides.

IMPORTANT: Methacrylates stored in bulk must be kept in contact with air (oxygen). So it will be desirable that the space is sealed by the gas that oxygen concentration is about 8%. This treatment may avoid the space going in flammable limits. The other side for avoiding polymerization, the oxygen concentration of the space needs suitable level. Monomer vapours are uninhibited and may form polymers in vent or flame arresters, resulting in blockage of vents.

Storage Temperature: Store at temperatures not exceeding 35°C. When not using immediately, it is preferable to store at temperature not exceeding 30°C.

Storage Life: Provided proper storage and handling procedures are followed, the product may be stored for up to 3 months from the date of receipt. If the product is to be placed in long-term storage (over three months), inspect the polymerization inhibitor concentrations for significant declines from initial concentrations as necessary. If polymerization happens and a container heats up, it should be transferred quickly to well ventilated wide area and cool down it by large quantity water.

Recommended Material: Steel drum, Plastic drum, Stainless drum and Steel 18L can.

8. Exposure Controls / Personal Protection

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Control concentration: Not established.

Exposure Limits: Not established.

Engineering Controls: If the product is used indoors, seal the point of emission or install ventilating equipment, or both. Install a safety shower, wash basin and eye bath near the handling area. Indicate the locations of these facilities clearly and prominently.

Personal Protective Equipment (PPE)

Respiratory Protection: Gas mask (to protect against organic gases), airline respirator (in atmospheres with high product vapor concentrations), air respirator, safety goggles or protective shield.

Hand Protection: Avoid skin contact. Wear oil-resistant gloves and boots.

Eye Protection: Wear chemical safety goggles and face shield.

Skin and body Protection: Additional protection may be necessary to prevent skin contact including use of apron, face shield, boots or full body protection.

Occupational Health Measures: Protective Equipment should be inspected periodically. Do not eat, drink or use tobacco when handling this product.

9. Physical and Chemical Properties

Appearance:	Clear, colorless liquid
Odor:	Weak ester odor
Flash point:	109°C(228.2 ° F) (open cup)
Auto-ignition Temperature:	N.A.
Boiling Point:	95°C(203 ° F) @ 13.3hPa
Melting Point:	<-60°C(-76 ° F)
Vapor Pressure:	N.A.
Vapor Density(Air=1):	4.5(air = 1)
% Solubility in Water:	∞
Specific Gravity:	1.072/@ 20°C
Viscosity:	6.79mPa · s/@ 20°C
pH:	N.A.
Molecular Weight:	130.15
Specific heat:	1.97 J/g · °C
Refractive index(nD²⁰):	1.4537(nD ²⁰)
SAPT:	>75°C

10. Stability and Reactivity

Stability: A polymerization inhibitor is added to the product to stabilize it under normal conditions of handling.

Hazardous Reactivity: Polymerization followed exothermic heat and may cause explosive reaction.

Condition to Avoid: Avoid to store far beyond the acceptable period or temperatures.

Incompatibility: Incompatible with strong acids and peroxide. (Cause of polymerization)

Hazardous Decomposition Products: N.A.

11. Toxicological Information

Acute Oral Toxicity:	LD ₅₀ (Rat) >4,000 mg/kg
Acute Dermal Toxicity:	LD ₅₀ (Rabbit) >3,000 mg/kg
Skin Corrosion/Irritation:	Eye Corrosion test by rabbit are reported that mild irritation is observed. PII: 0.5
Skin Sensitization:	Onset sample of allergic contact dermatitis is reported. Positive in human patch test. Maximization test by guinea pig is reported that it is recognized as a sensitizing substance.
Germ cell mutagenicity:	Negative in micronucleus test by rat bone-marrow cell
Carcinogenicity:	No information.
Reproductive Toxicity:	According to reproductive and developmental toxicity test by rat, this material did not show any genital toxicity or fertility toxicity at the dose of 1,000 mg/kg.
Target Organ Effects (Repeated):	According to repeated oral dose toxicity study by rat, urea nitrogen concentration and the weight of kidney is slightly increased when dose >30mg/kg or >100mg/kg.

12. Ecological Information

Organism Toxicity:	Pisces(cyprinodont) LC ₅₀ /96hr >100 mg/L
Persistence/degradability:	Readily biodegradable. Degradability by BOD is 95%.
Bioaccumulation:	Bioaccumulation is presumed to be low. log Kow= 0.47

13. Disposal Considerations

Follow the instructions given in Handling and Storage (Sec.7). Observe general precautions applicable to flammable toxic liquids.

When incinerating the product, soak up the product using sawdust or similar materials, and burn in small increments. When disposing the products, should entrust to qualified waste trader. We recommend that the Waste Water including these products are treated by activation sludge equipments and are wasted to public space after confirming the quality. When wasting the containers used these products, wash these by water after confirming that the containers do not remain the products.

14. Transport Information

International Regulations : Not hazardous by UN criteria.

UN Classification : Not applicable.

UN Number : N.A.

Container Classification : N.A.

Special Safety Measures : Avoid leak the products according to handle the container gently; do not overturn, drop, strike or drag. Watch the temperature of products, especially suddenly temperature raise.

Special Safety Measures and Conditions at Transportation : Follow the instructions given in Handling and Storage (Sec.7). Handle the container gently; do not overturn drop strike or drag. To be care accidental release and abnormal temperature raise of this container.

Proper Shipping Name: None

Non-Hazardous for Transport: This substance is considered to be non-hazardous for transport.

IATA (International Air Transport Association)

Non-Hazardous for Air Transport: Non-hazardous for air transport

15. Regulatory Information

Note: The latest local regulatory information should be confirmed with the importer.

16. Other Information

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Revision Indicator: Revised on 01/04/2021

Disclaimer: This SDS was prepared on the basis of materials, data and information available to us at the time of its writing. However, we make no warranty as to the accuracy of the data herein with respect to product content, physical and chemical properties thereof, or associated hazards. The precautions given herein apply to normal handling. For handling under special conditions, users are responsible for implementing adequate safety measures.